

Riko Miyagi

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EDUCATION

University of Waterloo: Bachelor of Environmental Studies, Geospatial Data Science (Co-op)

TECHNICAL SKILLS

Software/Tools: Excel, SQL, Python, R, Tableau, Power BI, BigQuery, SQLite, ArcGIS Pro, QGIS, SNAP, HTML, DrRacket, Microsoft & Google Ecosystem

EXPERIENCE

Energy Coordinator - [Ecobee](#), Toronto On Sep - Dec 2026

- Designed and developed an automated Power BI dashboard to streamline thermostat footprint count requests, reducing BigQuery query execution enabling faster response times across utility demand response programs.
- Conducted device analyses using BigQuery SQL to support utility partner program eligibility
- Prepared regular reports and dashboards on sales performance, CRM metrics, KPI achievement, providing actionable insights and recommendations to drive performance improvement

Energy Reporting and Measurement/Verification Assistant - [Parity Inc.](#), Toronto On Jan-Apr 2026

- Measured and verified energy and emissions savings from AI-driven HVAC optimization across 200+ commercial buildings in North America by collecting and analyzing building data.
- Produced energy savings reports including baseline comparisons and performance analyses to support guaranteed savings commitments for client portfolios.
- Detected and resolved anomalies in utility billing data across client portfolios; built automation tools in using AppScripts to streamline auditing workflows, reducing manual review time.

PROJECTS

[Heat Pump vs. Gas Furnace Feasibility Study](#) Feb 2026

- Developed a Python-based financial and environmental analysis tool comparing the 15-year total cost of ownership between air-source heat pumps and natural gas furnaces for Ontario homes.
- Modeled energy consumption using Ontario energy rates, and government rebates, finding a \$9,500 cost difference and 94.5% emissions reduction in favor of heat pumps environmentally.
- Produced visualizations and a professional report and prompted an interactive website showcasing results

[Ontario Electricity Demand Analysis](#) Apr 2026

- Cleaned and merged 12 months of IESO smart meter data in R to compare Residential and Small General Service (<50kW) electricity consumption patterns across Ontario
- Found opposing behavioral profiles due to different seasonal demand (residential winter heating vs. commercial summer cooling), informing targeted demand-response and conservation strategies.

[Site Suitability Analysis for a Small Wind Farm](#) July 2026

- Performed multi-criteria GIS analysis across Huron County, Ontario to identify two optimal wind farm sites using Weighted Overlay, Euclidean Distance modeling, and raster reclassification in ArcGIS Pro
- Delivered a client-facing report to the Ontario Government recommending sites with a combined 16 MW generation capacity, supported by professional map layouts

[Japan's Post-Fukushima Energy Transition \(2000-2024\)](#) Sept 2026

- Built an interactive Tableau dashboard analyzing Japan's electricity generation using U.S. EIA international energy data, visualizing the shift in fuel sources following the 2011 Fukushima accident.
- Found that nuclear's average share of generation fell from 26% pre-Fukushima to 3.7% post-Fukushima, while non-hydro renewable generation grew 4.6x over the same period.

CERTIFICATIONS

[University of Michigan Python Coursera Course](#) (Core Python programming) Aug 2025

[Google Data Analytics Professional Certificate](#) (Spreadsheets, SQL, R, Tableau) Aug 2025